RACE
Regular Analysis Through Continuous Exercise

Redox and Equivalent Concept : RACE-01

Introduction; Classical and Modern Idea of Oxidation and Reduction Reactions

1. Classify each of the following unbalanced half–reactions as either an oxidation or a reduction:

(a) $Cl_2(g) \longrightarrow Cl^-(aq.)$ (b) $S_2O_3^{2-}(aq.) \longrightarrow S_4O_6^{2-}(aq.)$

(c) $NH_3(g) \longrightarrow NO(g)$ (d) $C_2O_4^{2-}(aq.) \longrightarrow CO_2(g)$

(e) $Hg_2^{2+}(aq.) \longrightarrow Hg^{2+}(aq.)$ (f) $Pb_3O_4(s) \longrightarrow PbCl_2(aq.)$

CRX397

Redox Reactions in terms of Electron Transfer; Oxidation Number; Oxidising and Reducing Agents

2. In the reaction, $2Na_2S_2O_3 + I_2 = Na_2S_4O_6 + 2NaI$, I_2 acts as :

(A) reducing agent (B) oxidising agent

(C) oxidising as well as reducing agent (D) none of the above

CRX359

3. In which of the following compound, the oxidation state of sulphur is +7 ?

(A) $Na_2S_2O_8$ (B) $H_2S_2O_7$ (C) H_2SO_5 (D) None

CRX398

4. Oxidation number of Fe in Fe_3O_4 are :

(A) +2 and +3 (B) +1 and +2 (C) +1 and +3 (D) +2 and 5

CRX399

5. Which of the following compounds are arranged in increasing oxidation number of S :–

(A) $H_2SO_3, H_2S, H_2SO_4, H_2S_2O_3$ (B) $H_2S_2O_3, H_2SO_3, H_2S, H_2SO_4$

(C) $H_2S, H_2SO_3, H_2SO_4, H_2S_2O_3$ (D) $H_2S, H_2S_2O_3, H_2SO_3, H_2SO_4$

CRX386

6. The reaction $H_2S + H_2O_2 \longrightarrow S + 2H_2O$ manifests

(A) Oxidising action of H_2O_2 (B) Reducing nature of H_2O_2

(C) Acidic nature of H_2O_2 (D) Alkaline nature of H_2O_2

CRX387

7. Which of the following can only acts as oxidising agent?

(A) $KMnO_4$ (B) K_2MnO_4 (C) H_2O_2 (C) SO_2

CRX388

Disproportionation; Comproportionation; Intramolecular Redox

8. Which of the following is not a disproportionation reaction

(A) $2CHO.COOH + OH^- \rightarrow H_2C_2O_4 + HOCH_2.COOH$

(B) $Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$

(C) $2C_6H_5CHO \xrightarrow{Al(OC_2H_5)_3} C_6H_5COOCH_2C_6H_5$

(D) $4CrO_5 + 6H_2SO_4 \rightarrow 2Cr_2(SO_4)_3 + 6H_2O + 7O_2$ **CRX358**

ANSWER KEY

Question	1						2	3
Answer	(a) Reduction	(b) Oxidation	(c) Oxidation	(d) Oxidation	(e) Oxidation	(f) Reduction	B	D
Question	**4**	**5**	**6**	**7**	**8**			
Answer	A	D	A	A	D			

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Redox and Equivalent Concept : RACE-02

Equivalent Weight and Valency Factor (n-factor)

1. Equivalent weight of NH_3 in the change $N_2 \rightarrow NH_3$ is :

(A) $\frac{17}{6}$ (B) 17 (C) $\frac{17}{2}$ (D) $\frac{17}{3}$

CRX355

2. In the reaction, $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$, the eq. wt. of $Na_2S_2O_3$ is equal to its :

(A) Mol. wt. (B) Mol. wt./2 (C) 2 x Mol. wt. (D) Mol. wt./6

CRX356

3. The equivalent weight of $Na_2S_2O_3$ as reductant in the reaction,

$Na_2S_2O_3 + H_2O + Cl_2 \rightarrow Na_2SO_4 + 2HCl + S$ is :

(A) (Mol. wt.)/1 (B) (Mol. wt.)/2 (C) (Mol. wt.)/6 (D) (Mol. wt.)/8

CRX357

4. Moles of $K_2Cr_2O_7$ used to oxidise 1 mol $Fe_{0.92}O$ to Fe^{+3} are

(A) $\frac{0.92}{6}$ (B) $\frac{70}{92}\times\frac{1}{6}$ (C) $\frac{0.76}{6}$ (D) $\frac{70}{92}\times\frac{1}{3}$

CRX360

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Balancing of Redox Reactions by Oxidation Number Method

5. In the ionic reaction,

$xBrO_3^- + yCr^{3+} + zH_2O \rightarrow pBr_2 + qHCrO_4^- + rH^+$

{x, y, z, p, q, r, are in simplest whole number ratio}

Find the value of (x + y + z) till you get single digit answer :

CRX361

ANSWER KEY

Question	1	2	3	4	5
Answer	D	A	D	C	2

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Regular Analysis Through Continuous Exercise

Redox and Equivalent Concept : RACE-03

Gram Equivalents; Normality and Law of Equivalence

1. (i) 90 gram $H_2C_2O_4$ dissolve into 2 litre of solution. Find Normality of solutions.
(ii) How much amount of NaOH required to prepare 100 ml of 2N solution.
(iii) Determine the number of gram eq. of solute in 250 ml of 0.1N Na_2CO_3
(iv) Find Normality of 0.2M $Cr_2O_7^{2-}$, reduced into Cr^{3+}
(v) 49 gram H_2SO_4 dissolve in solution to make 10N solution. Find volume of solution.

CRX362

2. If instead of given amount of sodium thiosulphate, 2 moles of sodium thiosulphate along with 3 moles of $CuSO_4$ were taken initially. Then moles of sodium cuprothiosulphate formed is -
(A) 0 (B) 1 (C) 1.5 (D) 2

CRX337

3. How many mole of $FeSO_4$, $H_2C_2O_4$ and FeC_2O_4 are oxidised separately by one mole of $KMnO_4$ in acidic medium respectively :-
(A) 5 ; 5/2 ; 5/3 (B) 2/5 ; 2 ; 5 (C) 5/3 ; 3/5 ; 2/5 (D) 3/5 ; 2 ; 5

CRX389

4. How many gram of $KMnO_4$ are contained in 4 litre of 0.05 N solution. The $KMnO_4$ is to be used as an oxidant in acidic medium ?
(A) 1.58 g (B) 15.8 g (C) 6.32 g (D) 31.6 g

CRX390

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Redox Titration

5. What volume of 0.1 M H_2O_2 solution will be required to completely reduce 1 litre of 0.1 M $KMnO_4$ in acidic medium.
(A) 2500 ml (B) 500 ml (C) 1000 ml (D) 1200 ml

CRX391

6. 10 mole of ferric oxalate is oxidised by x mole of MnO_4^- in acidic medium. The value of 'x' is-
(A) 12 (B) 4 (C) 40 (D) 18

CRX392

7. In alkaline medium ClO_2 oxidises H_2O_2 to O_2 and itself gets reduced to Cl^-. How many moles of H_2O_2 are oxidised by 1 mole of ClO_2 ?
(A) 1 (B) 1.5 (C) 2.5 (D) 3.5

CRX393

Acid Base Titration

8. Problem based on Acid - Base titration
(i) 25 ml of N/10 NaOH solution exactly neutralises 20 ml of a monbasic acid solution find Normality of mono basic acid.
(ii) How many ml of 1 M H_2SO_4 acid solution is required to neutralise 10 ml of 1 M NaOH
(iii) 10 ml, 10 M H_2SO_4 solution, mixed with 100 ml of NaOH solution. If resultant solution is neutral then find Normality of NaOH solution.
(iv) How many moles of NaOH required to neutralise 3 moles of KHC_2O_4
(v) 45 gram $H_2C_2O_4$ was dissolved in water and made upto 500 ml. How much volume of this solution required to neutralise 120 ml of N/6 NaOH solution.

CRX363

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9. A sample of H_2SO_4 solution having density 1.96g/ml. If 50 ml of this solution is completely neutralize by 2M, 100ml NaOH solution. Wt % of H_2SO_4 in the solution is.

(A) 6 % (B) 12.5 % (C) 10 % (D) 2 %

CRX394

Iodimetric and Iodometric Titration

Paragraph for Q.10

632 g of sodium thiosulphate ($Na_2S_2O_3$) reacts with copper sulphate to form cupric thiosulphate which is reduced by sodium thiosulphate to give cuprous compound which is dissolved in excess of sodium thiosulphate to form a complex compound sodium cuprothiosulphate ($Na_4[Cu_6(S_2O_3)_5]$).

$CuSO_4 + Na_2S_2O_3 \rightarrow CuS_2O_3 + Na_2SO_4$

$2CuS_2O_3 + Na_2S_2O_3 \rightarrow Cu_2S_2O_3 + Na_2S_4O_6$

$3Cu_2S_2O_3 + 2Na_2S_2O_3 \rightarrow Na_4[Cu_6(S_2O_3)_5]$ (sodium cuprothiosulphate)

In this process, 0.2 mole of sodium cuprothiosulphate is formed. (O = 16, Na = 23, S = 32)

10. Moles of sodium thiosulphate reacted and unreacted after the reaction are respectively

(A) 3 & 2 (B) 2 & 3 (C) 2.2 & 1.8 (D) 1.8 & 2.2

CRX331

Back Titration

11. 4.3 gm of an alkane is burnt in sufficient oxygen. The CO_2 formed reacts completely with 300 ml, 2N NaOH solution producing Na_2CO_3. The alkane should be

(A) C_5H_{12} (B) C_6H_{12} (C) C_6H_{14} (D) C_5H_{10}

CRX395

12. An aq. solution of 0.5M $KMnO_4$ is divided into two parts. One part of it requires 125 ml of 1.5M aq. solution of oxalate ions in acidic medium, while another part requires 270 ml of 0.5M aq. solution of iodide ions in neutral medium which are converted into I_2 only. Calculate total volume (mL) of the initial $KMnO_4$ solution.

CRX396

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ANSWER KEY

Question	1					2	3	4	5	6
Answer	(i) 1N	(ii) 8 gram	(iii) 0.025	(iv) 1.2 N	(v) 100 ml	A	A	C	A	A
Question	**7**	**8**					**9**	**10**	**11**	**12**
Answer	C	(i) 0.125	(ii) 5	(iii) 2	(iv) 3	(v) 10	C	C	C	240

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